Executive Summary

Customer Churn Analysis Using Business Analytics

Project Overview

Customer retention is one of the most important factors for the success of any business. Acquiring new customers is generally more expensive than retaining existing ones. Therefore, understanding customer behavior and identifying the reasons behind customer churn can help organizations improve customer satisfaction, increase revenue, and reduce business losses.

This project focuses on analyzing customer churn using business analytics techniques. A dataset containing more than 500 customer records was analyzed to identify patterns related to customer retention and customer churn. The project includes data cleaning, exploratory data analysis (EDA), statistical analysis, visualization, hypothesis testing, and business recommendations. Python was used as the primary programming language along with libraries such as Pandas, NumPy, Matplotlib, and SciPy.

Business Problem

The company observed that a significant number of customers were discontinuing their services. Customer churn directly impacts revenue and increases customer acquisition costs. Management required a data-driven analysis to identify the factors influencing customer churn and to develop strategies for improving customer retention.

Project Objectives

The main objectives of this project were:

Analyze customer data to identify churn patterns.

Understand the impact of contract types on customer retention.

Examine the relationship between monthly charges and customer churn.

Evaluate customer payment preferences.

Perform statistical analysis and hypothesis testing.

Provide business recommendations based on analytical findings.

Dataset Description

The dataset contains customer information including:

- Customer ID
- Customer Tenure
- Monthly Charges
- Total Charges
- Contract Type
- Payment Method
- Paperless Billing
- Senior Citizen Status
- Customer Churn

The dataset contains more than 500 customer records, making it suitable for business analysis and statistical evaluation.

Data Analysis and Key Findings

The dataset was cleaned by removing duplicate records and handling missing values. Exploratory Data Analysis (EDA) was performed using multiple visualizations, including bar charts, pie charts, histograms, and correlation analysis.

The following key observations were identified:

Month-to-Month contracts showed higher customer churn compared to One-Year and Two-Year contracts.

Customers with higher monthly charges were more likely to discontinue the service.

Long-term customers generated higher total revenue.

Credit Card, Bank Transfer, and Electronic Check were the most frequently used payment methods.

Paperless Billing was widely adopted among customers.

Senior Citizen customers represented a smaller portion of the customer base but may require additional retention strategies.

Statistical hypothesis testing indicated that monthly charges have a significant relationship with customer churn, supporting the need for pricing and retention improvements.

Business Recommendations

Based on the analysis, the following recommendations are proposed:

Encourage customers to upgrade from Month-to-Month contracts to longer-term plans.

Introduce loyalty programs and retention offers for existing customers.

Monitor customers with high monthly charges and provide personalized discounts where appropriate.

Improve customer engagement through targeted communication and support services.

Promote digital payment methods for faster and more convenient transactions.

Continue expanding paperless billing to improve operational efficiency.

Implementing these recommendations can help reduce customer churn, improve customer satisfaction, and increase long-term profitability.

Expected Business Impact

The proposed recommendations are expected to generate the following business benefits:

- Increase customer retention.
- Reduce customer churn.
- Improve customer satisfaction.
- Increase customer lifetime value.
- Enhance revenue growth.
- Improve overall business decision-making using data analytics.